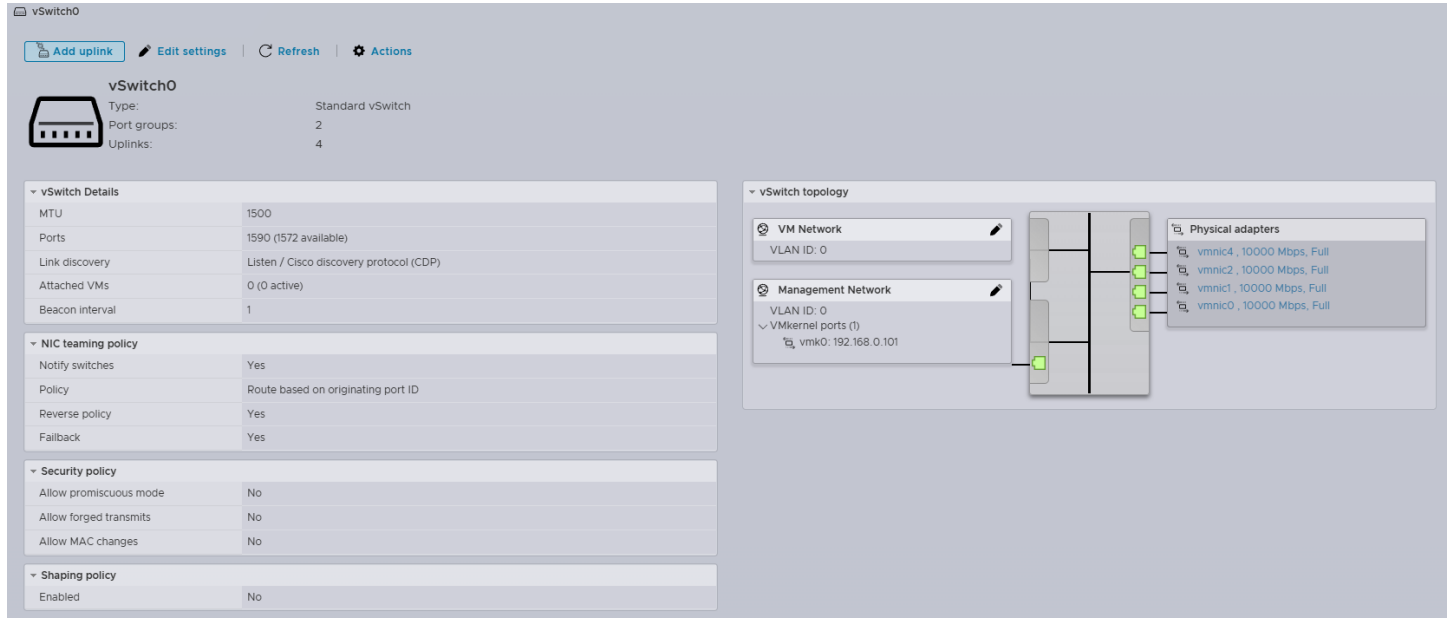
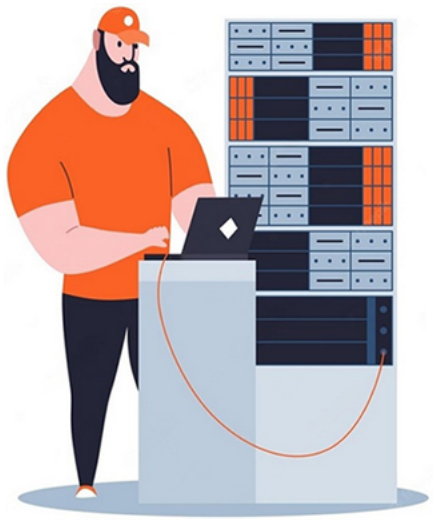
AGENDA

About NIC Teaming (vSS) in VMware ESXi 8.0

About Traffic Shaping Policy (vSS) in VMware ESXi 8.0





VMware ESXi
HYPERVISOR

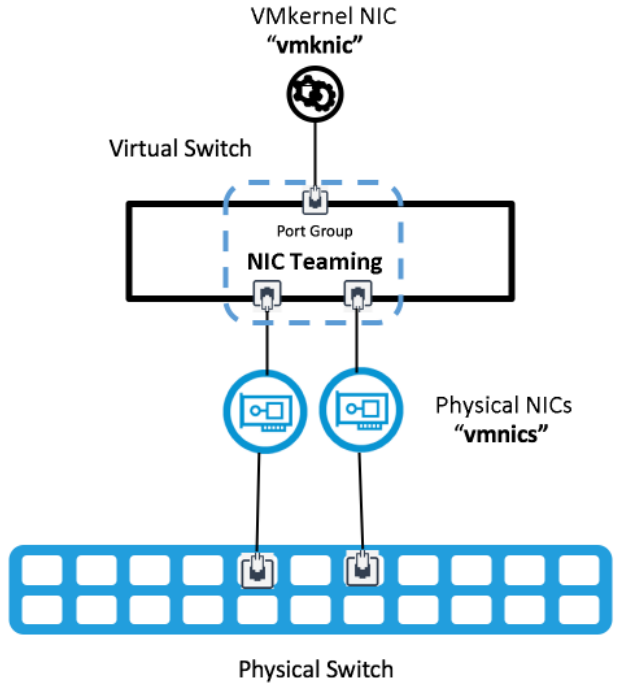
ABOUT NIC Teaming IN VMware ESXi 8.0



VMware ESXi
HYPERVISOR

NIC TEAMING IN VMware ESXi 8.0

NIC Teaming lets you increase the network capacity of a virtual switch by including two or more physical NICs in a team. To determine how the traffic is rerouted in case of adapter failure, you include physical NICs in a failover order. To determine how the virtual switch distributes the network traffic between the physical NICs in a team, you select load balancing algorithms depending on the needs and capabilities of your environment.



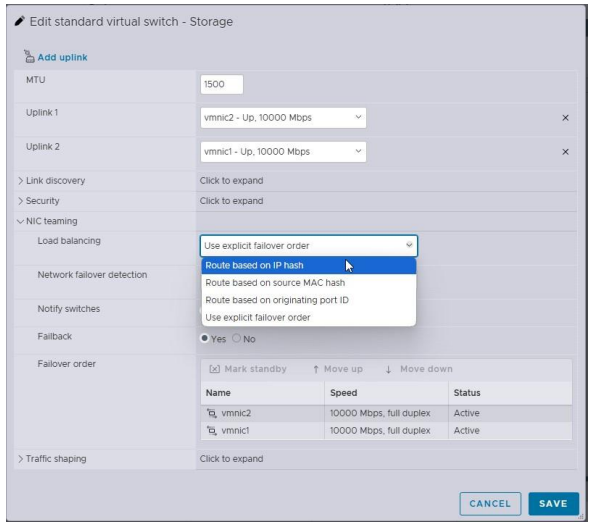


You can use NIC teaming to connect a virtual switch to multiple physical NICs on a host to increase the network bandwidth of the switch and to provide redundancy. A NIC team can distribute the traffic between its members and provide passive failover in case of adapter failure or network outage. You set NIC teaming policies at virtual switch or port group level for a vSphere Standard Switch and at a port group or port level for a vSphere Distributed Switch.

NIC TEAMING IN VMware ESXi 8.0

Load Balancing Policy

The Load Balancing policy determines how network traffic is distributed between the network adapters in a NIC team. vSphere virtual switches load balance only the outgoing traffic. Incoming traffic is controlled by the load balancing policy on the physical switch.





There are 4 type of NIC Teaming Load Balancing.

- Route based on the originating virtual port ID.
- Route based on IP hash.
- Route based on source MAC hash.
- Use explicit failover order.

NIC TEAMING IN VMware ESXi 8.0

Route based on the originating virtual port ID

Physical Layer Load Balancing. It always forwards traffic through the same uplink for this virtual machine or VMkernel adapter.

Route based on IP hash

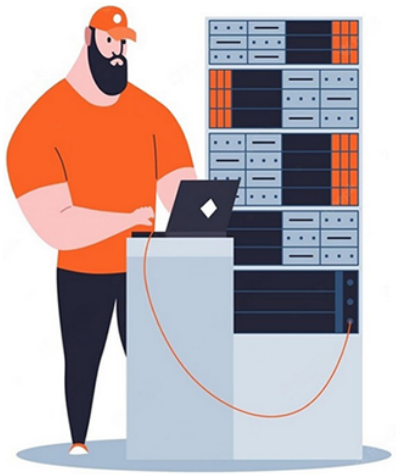
Network Layer Load Balancing, if first IP fails then second will up. Select an uplink based on a hash of the source and destination IP addresses of each packet. For non-IP packets, the switch uses the data at those fields to compute the hash.

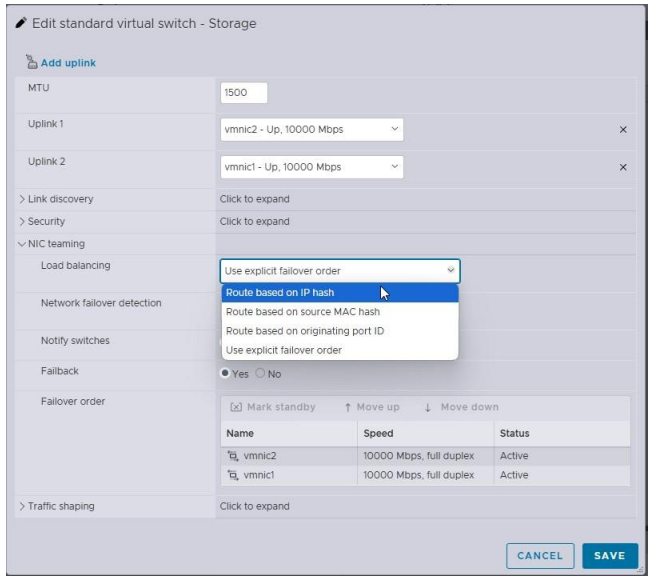
Route based on source MAC hash

Data Link Layer Load Balancing.

Use explicit failover order

Application Layer Load Balancing, It support user define rules, work with DRS. From the list of active adapters, always use the highest order uplink that passes failover detection criteria. No actual load balancing is performed with this option.





NIC TEAMING IN VMware ESXi 8.0

Network Failure Detection Policy

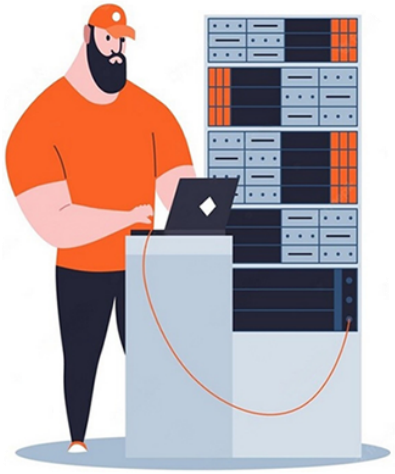
You can specify one of the following methods that a virtual switch uses for failover detection.

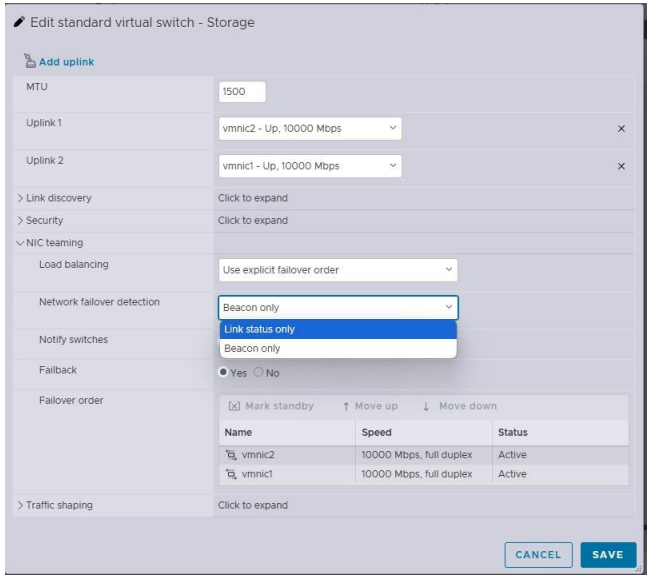
Link Status Only

- Relies only on the link status that the network adapter provides. Detects failures, such as removed cables and physical switch power failures. However, link status does not detect the following configuration errors:Physical switch port that is blocked by spanning tree or is misconfigured to the wrong VLAN .
- Pulled cable that connects a physical switch to another networking devices, for example, an upstream switch .

Beacon Probing

- This algorithm detect physical, network and MAC error. ESXi sends beacon packets every second.
- The NICs must be in an active/active or active/standby configuration because the NICs in an unused state do not participate in beacon probing.



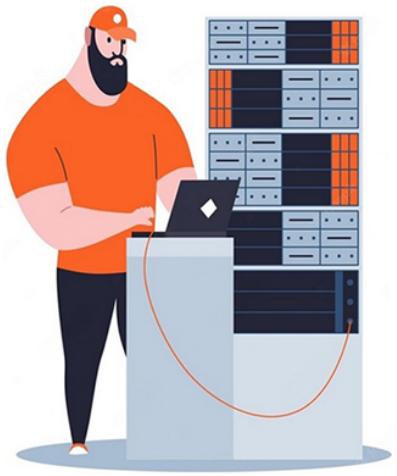


Both algorithm detect error in every 3 second minimum and 10 second maximum.

NIC TEAMING IN VMware ESXi 8.0

NIC Teaming – Override Switch Failover Order

- **Active Adapters**
  Continue to use the uplink if the network adapter connectivity is up and active.
- **Standby Adapters**
  Use this uplink if one of the active physical adapters is down.
- **Unused Adapters**
  Doesn't use this uplink in corporate level.



VMware ESXi
HYPERVISOR

ABOUT TRAFFIC SHAPING POLICY



VMware ESXi
HYPERVISOR

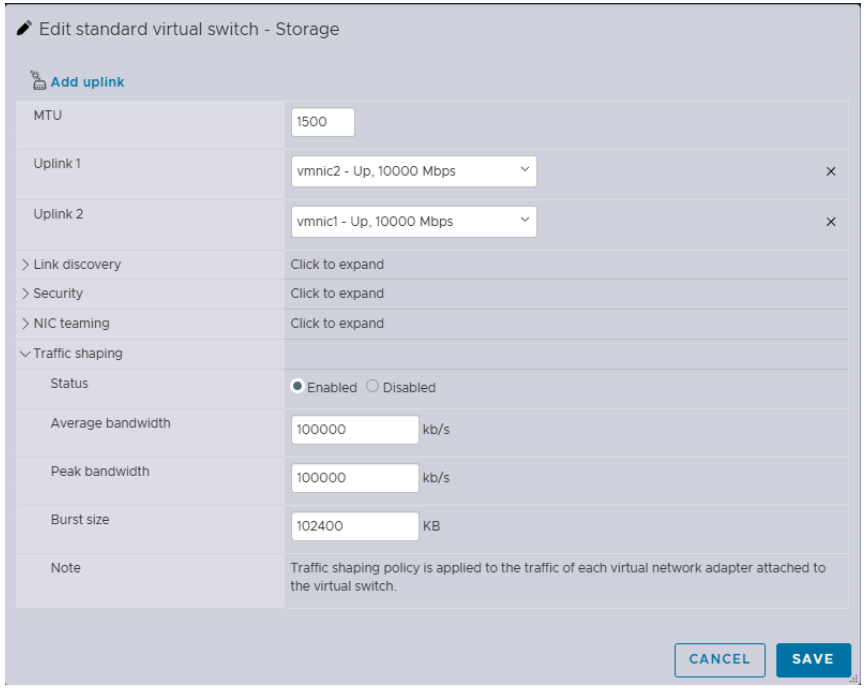
TRAFFIC SHAPING POLICY



Traffic Shaping Policy is the ability to control the quantity of traffic that is allowed to flow across a link. That is, rather than letting the traffic go as fast as it possibly can, you can set limits to how much traffic can be sent.

Traffic Shaping Policy is feature of vSwitch, it can be configured via vSphere Client or vSphere Web Client, by which you can manage bandwidth of VM clients.

Traffic Shaping Policy is defined by three characteristics:-



Average Bandwidth

The average amount of bandwidth, measured in kilobits per second (Kbps), that you allow the switch to send. There might be short periods where the traffic slightly exceeds this value, since it is an average over time, but for the most part, it will be enforced and traffic will go no faster than the defined speed limit set here. In daily bases you use this bandwidth for daily task such as “ping”.

Peak Bandwidth

The maximum amount of bandwidth that the switch is allowed to let through. The use of the peak bandwidth value is determined by how often we’ve hit the average bandwidth limitation. Whenever the actual traffic volume is lower than the average bandwidth limit, we gain what is called a “burst bonus” which can be any number of bytes up to the limit set by the burst size value. This bandwidth is useful for maximum server task such as Active Directory Authentication, HTTP, FTP, VPN, Name Resolving, etc.



Burst Bandwidth

This is an often misunderstood value, so we’ll go into detail. The burst size is the actual amount of “burstable” data that is allowed to be transmitted at the peak bandwidth rate in kilobytes. Think of the burst bonus as a network traffic savings account. And the burst size is the maximum number of bytes that can go into that account. So, when you need to send more traffic than the average bandwidth value allows, you transmit a burst of traffic, which is more than the allowed average bandwidth. But this burst, which always stays at or below the allowable peak bandwidth, will be forced to end when the number of bytes in your traffic savings account, your burst bonus, reaches zero. This bandwidth is total use of bandwidth or you can say Remaining bandwidth is called Burst bandwidth.